Individual Project

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1 Introduction

Tuberculosis (TB) remains a significant public health challenge in Brazil, with varying prevalence across regions influenced by socio-economic and environmental factors. This study employs Generalized Additive Models (GAMs) to analyze TB risk at a microregional level from 2012 to 2014, as GAMs provide flexibility in modeling non-linear relationships and capturing spatial and temporal effects. Unlike traditional regression models, GAMs use smooth functions to identify complex patterns in TB incidence rates concerning covariates like poverty, illiteracy, and urbanization. This approach improves predictive accuracy while accounting for regional and temporal variations. The study also incorporates residual diagnostics and spatial mapping to assess model adequacy and visualize TB risk distribution, offering insights for targeted public health interventions.

2 Data Understanding and Exploration

The TBdata dataset contains information on 557 microregions in Brazil from 2012 to 2014, focusing on tuberculosis cases and their potential socioeconomic and geographic determinants. It includes socioeconomic factors such as Indigenous population proportion, illiteracy, urbanization, dwelling density, poverty, sanitation quality, unemployment, and timeliness of TB case reporting. Additionally, it records the year, TB cases counts, population size, region ID and geographic coordinates i.e. Longitude and latitude for each microregion.

Preprocessing is essential to ensure the dataset is clean, consistent, and ready for analysis. The code 1 involves loading the dataset into R using the load function, making the data available for further manipulation. The TB rate per unit population is computed by dividing the number of TB cases by the population size for each microregion and year. This normalization allows for a meaningful comparison of TB incidence across regions with varying populations.

To maintain consistency, the column Region is renamed to COD_MICRO using `colnames(TBdata)[which(names(TBdata) == "Region")] <- "COD_MICRO"`. Standardized column names help streamline data manipulation and analysis.

Finally, missing values are checked after merging spatial data with predicted risk values. This is done using Code 9 ensuring data integrity and preventing biased results.[3][5]

3 Model Description and Fitting

A Generalized Additive Model (GAM) is used to capture non-linear relationships between TB incidence rates and socio-economic covariates. Traditional linear and generalized linear models (GLMs) assume fixed relationships between predictors and the response variable, which may not hold for complex epidemiological data. GAMs, on the other hand, allow for smooth, data-driven effects that can flexibly model these non-linear relationships, making them well-suited for TB risk assessment.

The general form of a GAM for Regression problems can be expressed as :

$$y_i = \beta_0 + \beta_1 x_i \ldots\ldots + \beta_p x_{ip} + \epsilon_i$$

The GAM includes smooth functions for socio-economic covariates and spatial coordinates, as well as factor for the year. [1][2]

The TB rate is modeled as a function of Indigenous population proportion, illiteracy, urbanization, population density, poverty, sanitation quality, unemployment, and timeliness of TB case notification, with spatial coordinates (longitude and latitude) and year as additional factors. The inclusion of s(lon, lat) accounts for spatial variation in TB risk, capturing regional patterns that might be influenced by environmental and healthcare access factors.

The model is fitted using the gam function from the mgcv package, applying the quasipoisson family to handle overdispersion[6]. A quasipoisson distribution is preferred over a normal distribution because TB counts exhibit overdispersion—where variance exceeds the mean—making standard Poisson regression inadequate. The use of Restricted Maximum Likelihood (REML) ensures optimal smoothness estimation, preventing overfitting while preserving the model's flexibility.[8]

Model diagnostics include residual analysis, deviance explained, and adjusted R-squared to assess fit quality. Predicted TB risk is mapped spatially to highlight high-risk regions, aiding in targeted public health interventions. These choices enhance the model's ability to uncover the complex interplay between socio-economic factors and TB incidence, supporting effective policy recommendations. The values in Figure 1 are calculated to provide a comprehensive assessment of the model's performance and its ability to explain the variability in TB rates.

4 Results

The model uses the quasipoisson family with a log link function. This choice is appropriate for count data with overdispersion, where the variance exceeds the mean[8].Table 1 presents the estimated significance levels (p-values) of various socio-economic and geographic factors affecting tuberculosis (TB) rates in Brazil. Each predictor variable is assessed to determine its statistical impact on TB incidence.

The diagnostic plot in Figure 2 as given in the Code 4 checks whether the residuals from the GAM model follow a normal distribution. The residuals generally align with the red reference line in the middle, indicating a reasonable approximation to normality. However, deviations in the tails suggest some degree of non-normality, which could be due to factors such as skewness or heavy tails. These deviations might indicate potential issues such as overdispersion or the presence of missing explanatory variables that could better account for the variability in TB incidence rates. [7]

The histogram in Figure 3 from Code 5 shows the distribution of residuals from a Generalized Additive Model (GAM). The residuals are mostly centered around zero, with a slight right skew, indicating that the model's predictions are generally accurate but may have some small positive errors. The frequency of residuals decreases as they move away from zero, suggesting that most predictions have minimal deviation. This plot helps assess whether the residuals follow a normal distribution, which is important for model evaluation.

The Figure 4 from Code 6 represent the results of a Generalized Additive Model (GAM) analyzing the relationship between various socio-economic and demographic factors with an outcome of interest. Each small plot illustrates the smooth function of a predictor variable, where the x-axis represents the predictor), and the y-axis shows the estimated smooth effect. The black line indicates the estimated function, while the gray shading represents the confidence interval. Some predictors, such as Indigenous and Unemployment, show a strong positive association, whereas others, like Illiteracy and Poor Sanitation, exhibit a negative or non-linear effect. The bottom spatial plot visualizes the geographic variation of the response variable, where contour levels and color gradients indicate regions of higher or lower effects. The red and orange areas suggest higher estimated values, while blue and yellow areas indicate lower values. This analysis highlights the significant role of socio-economic variables and geographical location in influencing the studied outcome.[2]

This boxplot in Figure 5 as a result of Code 7 illustrates the distribution of TB rates for each year from 2012 to 2014. The median TB rate appears relatively stable across the three years, suggesting no drastic changes in central tendency. However, the presence of numerous outliers, represented by dots above the whiskers, indicates that certain microregions experience significantly higher TB rates compared to the majority. This highlights a high variability in TB incidence across different regions, suggesting that localized factors may be influencing the spread of the disease. Understanding these variations is crucial for targeted public health interventions and resource allocation.[4]

The map in Figure 6 from Code 10 presents the predicted risk of tuberculosis (TB) by region in Brazil from 2012 to 2014. The risk levels are color-coded, with darker shades of purple representing regions with a higher predicted risk of TB and lighter shades indicating lower risk areas. The spatial distribution reveals that certain regions, particularly in the North and parts of the Northeast and South, exhibit higher TB risk, while central and eastern areas tend to have lower risk. This visualization likely reflects socio-economic, environmental, and healthcare access disparities across the country. The darker regions might correspond to areas with higher population density, poorer sanitation, or greater poverty—factors commonly associated with TB prevalence.

References

[1] Generalized Additive Models Using R - GeeksforGeeks

[2]Intro to GAMs - Bristol University

[3] (PDF) Estimation of underreporting in Brazilian tuberculosis data, 2012-2014

[4] The R Graph Gallery – Help and inspiration for R charts

[5] ChatGPT

[6] Using random effects in GAMs with mgcv

[7] QQ Plot: Uses, Benefits & Interpreting - Statistics By Jim

[8] R Handbook: p-values and R-square Values for Models

Appendix A: Plots and Tables

Predictor Variable	P - value	Description
Indigenous	1.04×10^{-6}	Has a strong effect on TB rates
Illiteracy	0.182151	Do not have a significant effect on TB rates
Urbanisation	6.86×10^{-5}	Has a significant effect on TB rates
Density	6.31×10^{-5}	Influences the TB rates
Poor Sanitation	$<2 \times 10^{-16}$	Highly significant i.e. Strongly affects TB rates
Unemployment	$<2 \times 10^{-16}$	Highly significant i.e. Strongly affects the TB rates
Poverty	0.000157	Significant effect on TB rates
Timeliness	$< 2 \times 10^{-16}$	Timeliness of TB cases notification affects TB rates
Longitude and latitude	$< 2 \times 10^{-16}$	Significant effect on TB rates

Table 1. Approximate Significance of Smooth Terms

```
R-sq.(adj) =  0.484   Deviance explained = 52.5%
-REML = -7526.9  Scale est. = 4.6017e-05  n = 1671
```

Figure 1

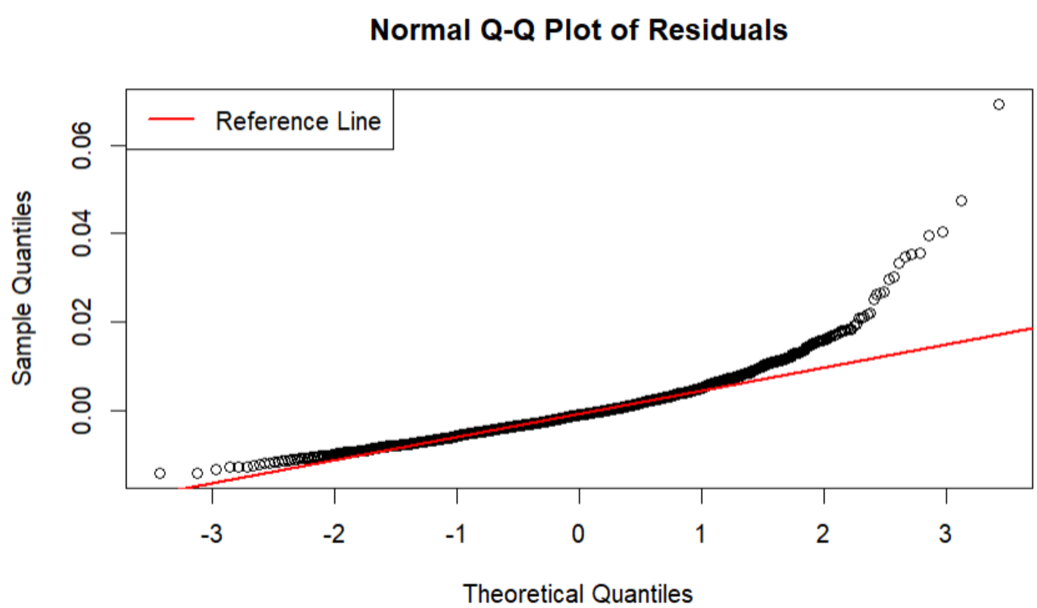

Figure 2

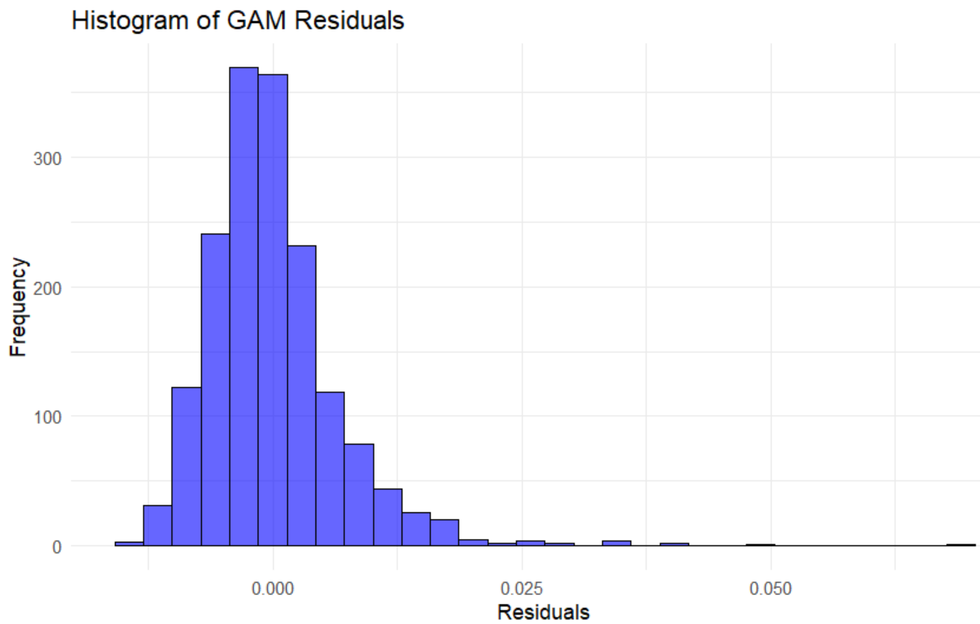

Figure 3

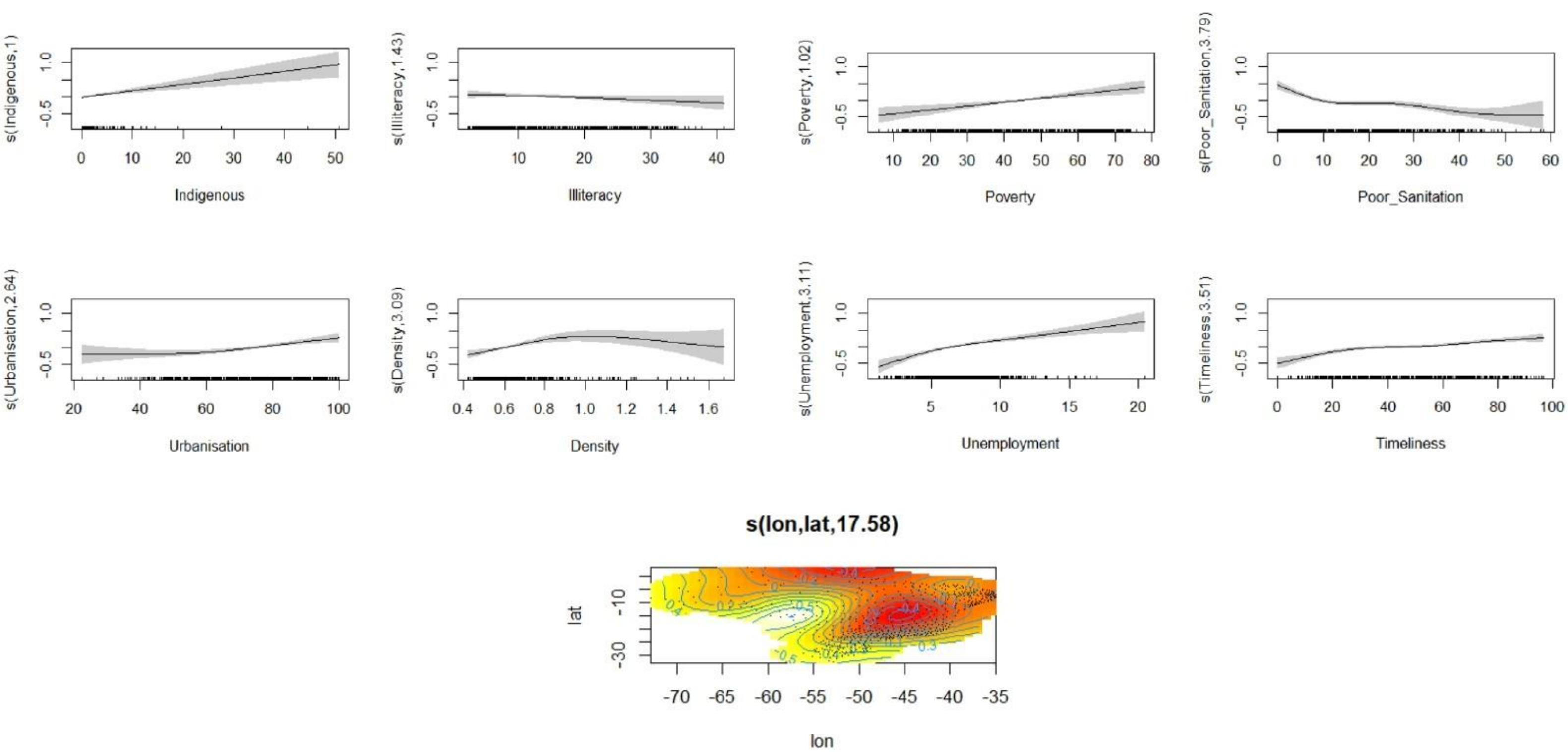

Figure 4 – Smooth Effects of Variables

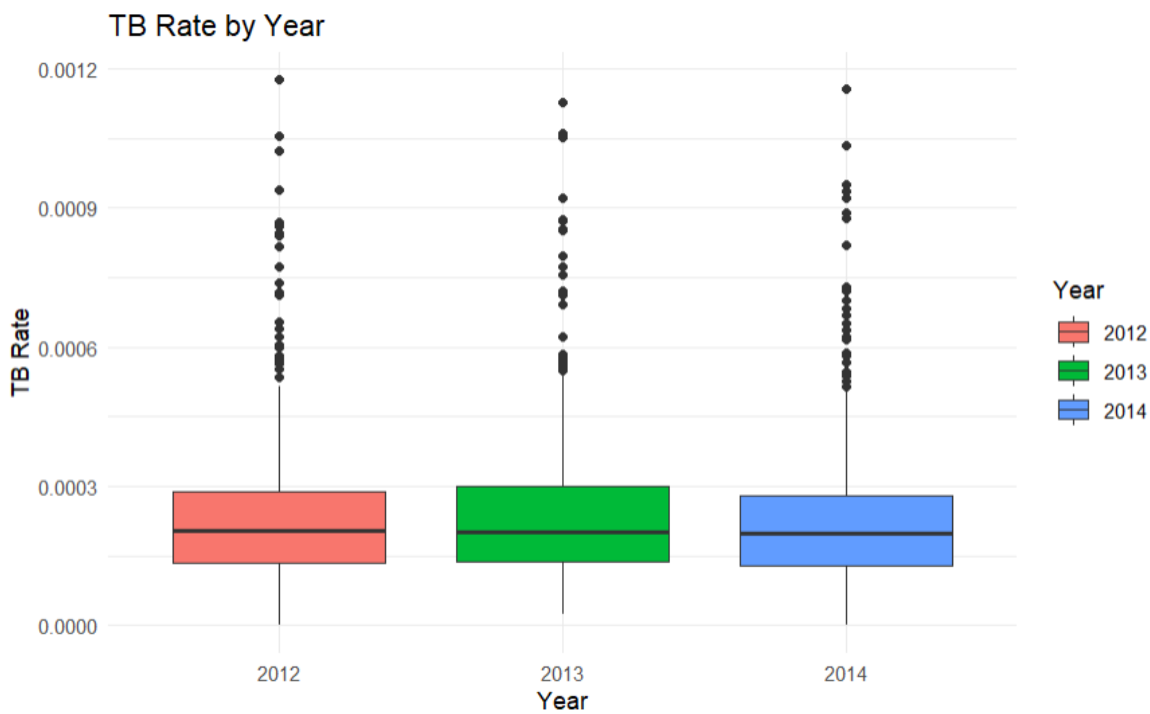

Figure 5

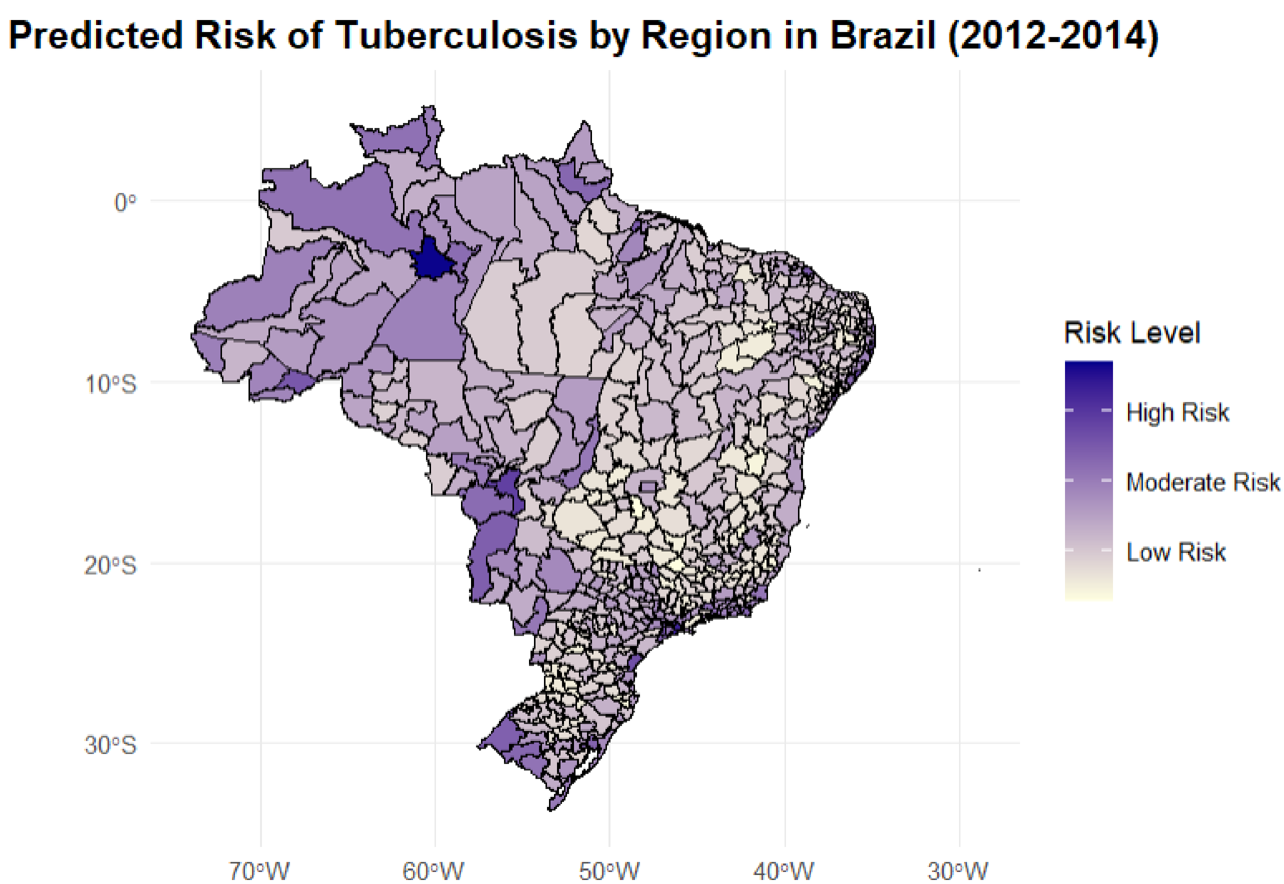

Figure 6

Appendix B: Raw Code

Code 1 : Load necessary libraries and load the dataset

```
library(mgcv)
library(sp)
library(maps)
library(fields)
library(sf)
library(ggplot2)
library(dplyr)
library(viridis)
load("datasets_project.RData")
```

Code 2 : Fit a Generalized Additive Model (GAM) with spatial and temporal effects

```
gam_model <- gam(TB_rate ~ s(Indigenous, k=5) + s(Illiteracy, k=5) +
                 s(Urbanisation, k=5) + s(Density, k=5) + s(Poverty, k=5) +
                 s(Poor_Sanitation, k=5) + s(Unemployment, k=5) +
                 s(Timeliness, k=5) + factor(Year) + s(lon, lat, k=20),
                family = quasipoisson, data = TBdata, method = "REML")
```

Code 3: Predict TB risk

```
TBdata$Predicted_Risk <- predict(gam_model, type="response")
residuals_gam <- residuals(gam_model, type = "pearson")
```

Code 4: Normal Q-Q Plot of Residuals

```
qqnorm(residuals_gam, main = "Normal Q-Q Plot of Residuals")
qqline(residuals_gam, col = "red", lwd = 2)
legend("topleft", legend = "Reference Line", col = "red", lwd = 2)
```

Code 5: Histogram of Residuals

```
ggplot(data = data.frame(residuals_gam), aes(x = residuals_gam)) +
```

```
  geom_histogram(bins = 30, color = "black", fill = "blue", alpha = 0.6) +
  labs(title = "Histogram of GAM Residuals", x = "Residuals", y = "Frequency") +
  theme_minimal() +
  theme(legend.position = "right")
```

Code 6: Smooth Effects of Variables

```
plot(gam_model, pages = 3, scheme = 2, rug = TRUE, seWithMean = TRUE, shade = TRUE)
legend("topright", legend = "Smooth Terms", col = "black", lwd = 2)
```

Code 7: TB Rate by Year

```
ggplot(TBdata, aes(x = factor(Year), y = TB_rate, fill = factor(Year))) +
  geom_boxplot() +
  labs(title = "TB Rate by Year", x = "Year", y = "TB Rate", fill = "Year") +
  theme_minimal() +
  theme(legend.position = "right")
```

Code 8: # Convert brasil_micro to an sf object (spatial data) and merge with predicted risk

```
brasil_micro <- st_as_sf(brasil_micro)
merged_data <- merge(brasil_micro, TBdata, by = "COD_MICRO", all.x = TRUE)
```

Code 9: Check for missing data after merging

```
missing_values <- sum(is.na(merged_data$Predicted_Risk))
print(paste("Missing values after merging:", missing_values))
```

Code 10: Plot the map of Predicted Risk using ggplot and print the map

```
plot_tb <- ggplot(data = merged_data) +
  geom_sf(aes(fill = Predicted_Risk), color = "black", size = 0.1) +
  scale_fill_gradient(name = "Risk Level",
            low = "lightyellow", high = "darkblue",
            breaks = c(0.0002, 0.0004, 0.0006),
            labels = c("Low Risk", "Moderate Risk", "High Risk")) +
```

```
  labs(title = "Predicted Risk of Tuberculosis by Region in Brazil (2012-2014)") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, size = 14, face = "bold"),
        legend.position = "right")
print(plot_tb)
```

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